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Triboelectric nanogenerator based on reactivated electrode materials derived from waste alkaline battery: Influence of pyrolysis temperature and surface morphology

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ABSTRACT

Triboelectric nanogenerators (TENGs) have been extensively studied in recent years due to their potential for energy harvesting and self-powered sensing. However, most of the research has focused on only exploring new materials, while there has been limited examination of how optimizing material properties or device system designs could enhance the TENG devices' performance. To address this gap, this research presents a pioneering approach using thermally reactivated electrode materials derived from waste alkaline batteries (WABs) for future energy harvesting applications in TENGs. By systematically investigating how the oxygen content and surface morphology affect TENG device performance at different pyrolysis temperatures, we achieved optimal TENG material at 500 °C, exhibiting outstanding output voltage, current, and power density of 790 V, 40.27 μA, and 529.375 μW/cm^2, respectively. Mechanical stability testing revealed that the TENG device based on RC-WABs remains stable for over 40,000 cycles with consistent output performance. Furthermore, the proposed method employs a simple and efficient pyrolysis technique to reactivate the cathode and anode materials of WABs, operating at low temperatures without activating gases or inert atmospheres. To validate real-time applicability, the TENG device was successfully employed to power various applications, such as illuminating 150 LEDs and operating a digital calculator. This research provides a new perspective on the use of waste materials for energy harvesting applications and offers a potential solution for recycling waste alkaline batteries, which is one of the significant environmental problems. Overall, the findings highlight the potential of the proposed approach for future TENG device design and provide a new direction for sustainable energy harvesting.

1. Introduction

Nowadays, the proliferation of electronic gadgets has revolutionized human activity, making it heavily reliant on electronic devices [1–3]. However, this gadget-dependent world has given rise to the pressing issue of electronic waste (e-waste) generation, posing significant environmental and health challenges [4,5]. This multi-faceted issue encompasses all unusable electronic devices and their subcomponents, presenting a significant challenge for our environment. Batteries emerge as a paramount concern within the vast array of e-waste contributors, with alkaline batteries (ABs) reigning supreme. Their ubiquitous presence in a myriad of electronic devices, ranging from cell phones and toys to sports equipment and medical devices, accentuates their pivotal role in fueling the e-waste crisis [6–9].

People typically throw ABs away as waste once they have reached the end of their useful life. Unfortunately, ABs are commonly disposed of alongside other household waste, such as food and plastic waste [10–13]. It is well known that the primary components of ABs consist of fine-grained manganese dioxide (MnO_2) combined with carbon dust as the cathode (70%), zinc chloride (2–10%) as the anode, and potassium hydroxide (KOH) as the electrolyte (7–10%) [14–16]. Consequently, it has a significant impact on the health of living organisms and the

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environment [7,17–19]. It has already been reported that metal ions like Mn (cathode), Zn, and Cl_2 (anode) and flammable and poisonous organic electrolytes are harmful to humans when they enter the body [20–23]. Mainly, they entered through food, air, and water. Exposure to high levels of these metal ions, either through ingestion or inhalation, may lead to various health issues. When ABs are incinerated as part of waste disposal, toxic gases such as MnO_2 and $ZnCl_2$ are released into the air. These emissions can have detrimental effects on the lungs and central nervous system, potentially increasing the risk of neurological disorders, bronchitis, and pneumonitis [17,24,25]. Furthermore, they can cause harm to the skin and eyes, as the resulting aerosols are highly irritating to the respiratory tract [17,26]. The chemicals present in ABs also contaminate soil, groundwater, and surface water. Metal ions, denser than water, exhibit toxic or poisonous behavior even at low concentrations (ppm level) [27]. Previous studies have demonstrated that overexposure to metal ions is associated with brain chemical balance disruptions. Specifically, when these ions affect the dopaminergic system, they may potentially trigger Parkinson's disease, which can be fatal [28–31]. In conclusion, the improper disposal of ABs and the release of their toxic components have severe consequences for both human health and the environment. Adopting proper disposal methods and promoting the recycling of these batteries are crucial for mitigating these risks.

With the escalating global concern regarding proper waste management, effectively handling waste materials has become a pressing priority. The limited capacity of landfills to accommodate waste necessitates alternative solutions, with recycling emerging as an integral approach in recent years. While recycling waste alkaline batteries (WABs) has been explored in various applications such as energy conversion [32,33], storage [33], sensing [34], and environmental cleanup [12], their potential in triboelectric nanogenerator (TENG) remains unexplored. Our paper presents a new material for TENG applications that is cost-effective, scalable, and derived from waste alkaline batteries (WABs). We acknowledge the critical need for an economical and readily available counter-electroactive layer, and through reactivating the cathode and anode materials derived from WABs (RC-WABs and RA-WABs), we introduce RC-WABs as a viable solution for TENG technology for the very first time [35,36]. This marks the first instance of its utilization in this context. Our experimental investigations demonstrate that TENG devices utilizing the reactivated cathode of waste alkaline batteries (RC-WABs) outperform those employing the reactivated anode material of waste alkaline batteries (RA-WABs). This research makes a significant contribution to waste management and renewable energy research. Furthermore, by proposing the utilization of RC-WABs as an economic counter-electroactive layer in TENG devices, we offer a sustainable solution to waste management while showcasing the remarkable performance potential of WABs in energy harvesting applications. The findings of our study underscore the economic and environmental benefits associated with the large-scale implementation of WABs in TENG technology. We believe this research offers a promising direction for the future of sustainable energy.

Currently, various strategies, including hydrometallurgical [37,38], pyrometallurgical [39,40], and direct physical method [41–43], are applied to reactivate or recycle the electrode material as well as metal components of batteries. Despite efforts to reuse and upcycle e-waste, there is still an urgent need for simple, economical, environmentally friendly, and energy-saving approaches. However, in this study, we used the pyrometallurgical approach to reactivate both the electrodes (cathode and anode) of WABs due to its low cost, ease of use, and high level of productivity [44–46]. It is worth noting that our synthesis mainly avoided the inert atmosphere and operated at low temperatures without the use of activating gases (N_2, H_2, Ar, etc). In order to achieve the extraordinary material needed to harvest more and more energy through TENG, the optimal pyrolysis temperature has been thoroughly investigated. For that, we altered the reaction temperatures ranging from 0, 100, 200, 300, 400, 500, 600, and 800 °C and made several TENG devices, viz., RC-WABs-x (where x = 0, 100, 200, 300, 400, 500, 600, and 800 °C, which mean activation temperatures), based on their reaction temperatures. The results showed that a reactivating temperature of 500 °C is feasible for achieving efficient tribopositive friction layered material derived from the cathode material of WABs. More importantly, as compared to TENG devices based on RA-WABs, the TENG device based on RC-WAB operate exceptionally well at low temperatures (500 °C) and it was found to be notable output voltage, current, and power density of 790 V, 42 μA, and 529.375 $\mu W/cm^2$, respectively. Hence this paper mainly focused on cathode reactivation which might provide potential material for future TENG technology. It was also tested for mechanical stability and was found to be stable for more than 40,000 cycles with stable output performances. More importantly, the proposed TENG device RC-WABs-500, generated enough power to run 150 LEDs and a calculator. Overall, the results of the experiment showed that the RC-WAB-500 tribopositive material is more stable and effective than waste-based TENG devices that were reported earlier.

Until now, significant efforts have been dedicated to achieving high output efficiency in TENG devices [47,48]. These endeavors have explored various approaches, including the utilization of novel nanomaterials [49], two- and three-dimensional (2D and 3D) materials [50–53], enhancing surface conductivity [54–57], fabricating heterostructures and composite devices [58,59], controlling device thicknesses [48,60,61] and so on. However, the impact of oxygen content and surface morphology of the electroactive layer on TENG performance remains largely unexplored because of the lack of comprehensive guidelines in this area. To address this knowledge gap, we systematically investigated the influence of surface morphology and oxygen content on the performance of the TENG device made by using a reactivated cathode material of WABs as a tribopositive layer. Our study employed various characterization techniques, including X-ray diffraction (XRD), field-emission scanning electron microscopy (FE-SEM), and Fourier-transform infrared spectroscopy (FTIR), to establish the relationship between pyrolysis temperature, structural characteristics, and device performance.

This research contributes two notable aspects to the field of TENG energy harvesting. Firstly, we successfully repurpose the cathode material of WABs as a tribopositive layered material for TENG applications through a pyrometallurgical approach. Additionally, we thoroughly examine the impact of surface morphology and oxygen content of the electroactive layered material on TENG performance. By investigating these crucial factors, we provide valuable insights for sustainable WAB recycling and the fabrication of promising materials for TENGs. The findings of this study hold great significance for the advancement of TENG technology. The comprehensive understanding of the relationship between surface morphology, oxygen content, and TENG performance offers opportunities for optimizing TENG devices and enhancing their overall efficiency. The proposed work not only contributes to the recycling of waste alkaline batteries but also paves the way for the development of efficient and environmentally friendly materials for TENG energy harvesting, fostering sustainable progress in this burgeoning field.

2. Materials and experiment

2.1. Reagent and materials

The waste alkaline batteries (WABs) were gathered from the university's electronic waste disposal unit, Jeju National University, South Korea. Film Bank, a Korean company, provided Polytetrafluoroethylene (PTFE), Polyethylene Terephthalate (PET), and Kapton. From Elephant's electronic shopping website, copper wires, aluminum, and copper tapes were purchased.

2.2. Preparation of RC-WABs tribopositive materials

To create value-added products from waste materials, we re-engineered, rebooted, and refilled the cathode material from waste

alkaline batteries (WABs) to harvest electrical energy from external mechanical energy through TENG. The WABs were typically collected from the electronic waste disposal unit, at Jeju National University. The first step involved precisely separating or dismantling the cathode, which was tightly packed in the shell. To make a fine powder, the dismantled cathode source was manually ground with a mortar and pestle. A simple, easy, and straightforward pyrolysis method was used for the reactivation of cathode material. In a typical procedure, 5 g of fine cathode material was first transferred into a silica crucible. This crucible was then wrapped in aluminum foil and placed in a muffle furnace. It was kept at 500 °C for 3 h in the absence of any activating gases such as N_2, H_2, or Ar (heating rate: 3 °C/min). However, to achieve different morphologies of RC-WABs, we varied the reaction temperature ranging from 0 to 800 °C. Furthermore, after reaction completion, the furnace was allowed to cool naturally to room temperature. The obtained material was then transferred to a mortar and pestle and crushed into a fine powder, which was then labeled as reactivated cathode material from waste alkaline batteries (RC-WABs). Following that, the RC-WABs were stored in a separate glass vial to identify specific RC-WABs based on their reaction temperature ratios (0 to 800 °C). It was then tested as a positive layered material for TENG research throughout the experiment for the first time. A typical RC-WAB synthesis route is depicted in Fig. 1**a.**

2.3. Preparation of RA-WABs tribopositive material

A process similar to the one used for making tribopositive layered material, RC-WABs, was employed to reactivate the anode material of WABs (RA-WABs). The preparation step is replacing the cathode with an anode as the reaction precursor material while keeping the other reaction parameters (reaction instrument, reaction environment, reaction temperature, reaction time) constant. The furnace was allowed to cool naturally to room temperature after the reaction was completed. The obtained material was then crushed into a fine powder using a mortar and pestle and labeled as RA-WABs. The RA-WABs were then placed in a glass vial for further testing. It was then tested as a tribopositive layered material and then as a tribonegative layered material.

2.4. Fabrication and mechanism of RC-WABs-based TENG device

The typical construction of an RC-WABs-based TENG device is depicted in Fig. 1**b.** In this work, the investigated RC-WABs material was used as the tribopositive layer while the PTFE was used as the tribonegative layer as shown in Fig. 1**b**. The top and bottom electrodes were made with copper (Cu) and aluminum (Al) conductive adhesive tapes, respectively, and supported by PET sheets that were flexible on both sides, as shown in Fig. 1**b**. The triboelectric layers are supported by the PET sheet, allowing the manufactured device to press and release quickly at a specific frequency as shown in Fig. 1**b**. After numerous back-to-back experiments, the TENG device's dimensions were precisely adjusted to 4 × 4 cm^2, with copper lead attached to the underside of both Al and Cu tapes. Fig. S1**(a)**, Fig. S1**(b), and** Fig. S1**(c)** indicates the top view, front view, and side view of the as-fabricated RC-WABs TENG device respectively. While Fig. S1**(d)** pressing state of the device.

The PTFE was utilized as a tribonegative layer whereas the developed RC-WABs were used as a tribopositive layer in the investigated RC -WABs-based TENGs, as shown in Fig. 1**c**. Fig. 1**c(i)** depicts the contact between two materials, namely the tribopositive layer RC-WABs and the

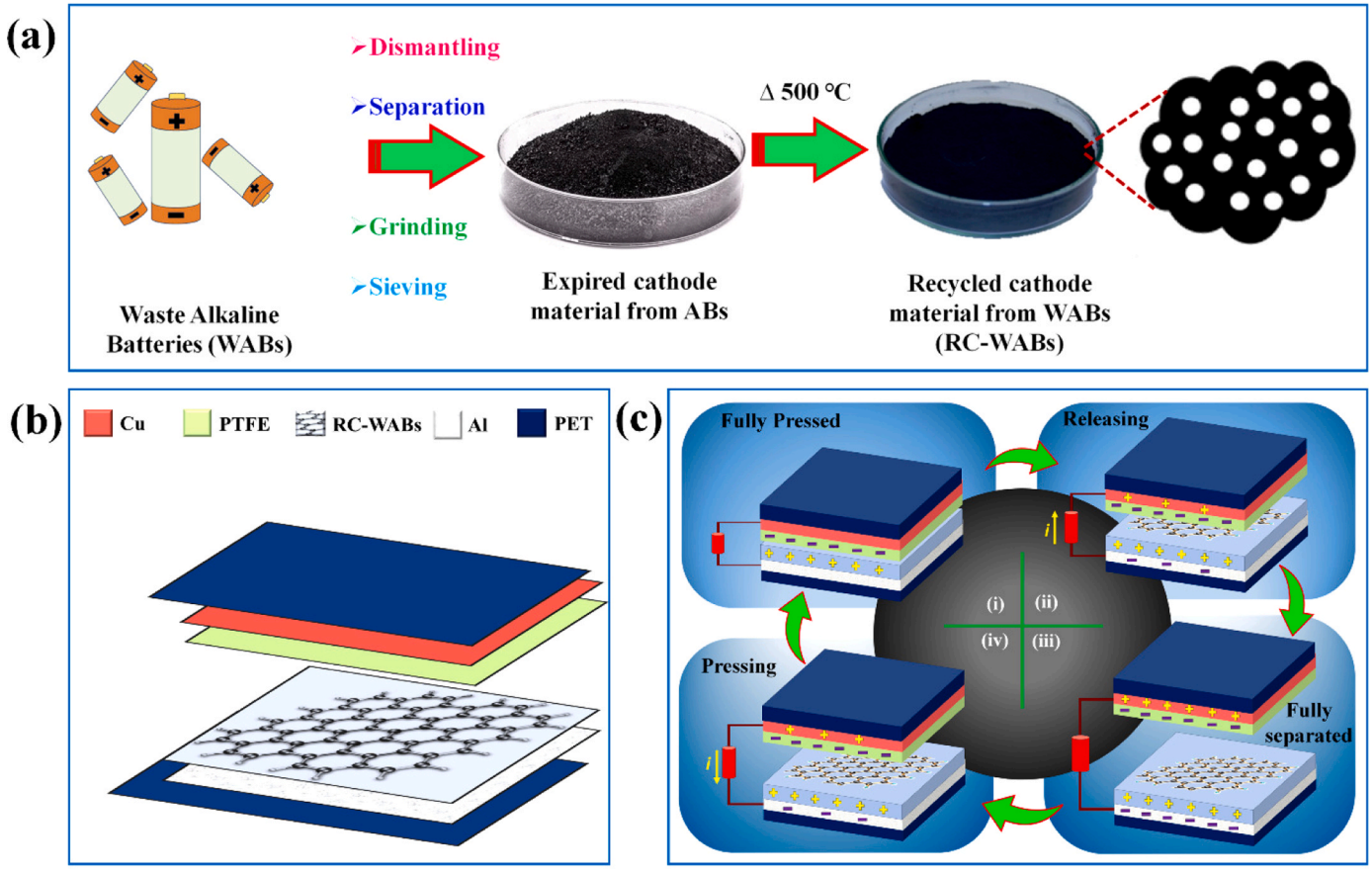


Fig. 1. : (a) Schematic representation of cathode reactivation of waste alkaline batteries (RC-WABs). (b) Graphical structure for the fabrication of RC-WABs-based TENG. (c) Schematic illustration of the working mechanism of the RC-WABs-based TENG device, (i) fully pressed: the electric charges are generated on PTFE and RC-WABs, (ii) releasing: electrons flow from one electrode to another, (iii) fully separated: no current flows, and (iv) again pressing: electrons again flow from one electrode to another electrode.

tribonegative layer PTFE when an external pressure is applied. The RC-WABs device facilitates the transfer of electric charge between these two materials due to their distinct electron energies. Consequently, the interface of the RC-WABs layer becomes positively charged, while the interface of the PTFE layer becomes negatively charged. When an external force is released, an alternating current is established through the circuit. This occurs because the charge generated on both electrodes and the presence of free electrons cause them to move from one electrode to the opposite electrode, as depicted in Fig. 1**c(ii).** The movement of electrons continues until the force acting upon them is wholly separated, as shown in Fig. 1**c(iii),** at which point the current ceases. Subsequently, when the tribopositive and tribonegative layers come into contact again due to the influence of an external force, current flows through the external circuit. This occurs because the previously excited energy flows back, as Fig. 1**c(iv)** illustrates. As a result, this cyclic process continues, generating a continuous current. In more scientific terms, the phenomenon described in Fig. 1**c** is an example of the triboelectric effect, where two materials with different electron affinities (RC-WABs and PTFE) experience charge transfer when they come into contact and then separate. This charge transfer is attributed to the difference in electron energies between the materials, leading to the formation of positive and negative charges at their respective interfaces. When an external force is applied to bring the materials into contact, an electric current is induced due to the movement of charge carriers, such as electrons, through an external circuit. The separation of the materials interrupts the current flow. However, upon their recontact, the previously separated charges are able to flow again, resulting in the generation of a current. This cyclic process of charge separation and recombination, driven by the triboelectric effect, enables continuously generating an alternating current. This phenomenon finds applications in various devices and technologies, such as triboelectric generators, where mechanical energy is converted into electrical energy through the repeated contact and separation of different materials.

2.5. Characterizations and measurements

Using an oscilloscope (DSO6052A, Agilent Technologies) and a source/measurement instrument (B2902A, KEYSIGHT), respectively, the output voltage, current, and transferred charge were observed to ascertain the electrical output capabilities. A typical linear mechanical motor (SHINYEONG MECHATRONICS CO., LTD., C32B 16–50) has been employed to apply external forces corresponding to stretching and releasing actions. A Panalytical X'PERT PRO X-ray diffraction microscope was used to examine the RC-WABs under study to ascertain their phase structure, crystallinity, and composition. The morphological traits of the RC-WABs (TESCON MIRA 3) were investigated using field emission scanning electron microscopy. The Fourier-transform infrared spectroscopic analysis was carried out using an ALPHA II Fourier-transform infrared spectrophotometer (4000–450 cm^{-1}). The power was calculated using the equation $P = I^2R$, where I stand for current and R for external load resistance.

3. Results and discussion

3.1. Characterization

3.1.1. FESEM-EDS analysis

The previous research has highlighted the influence of material morphology on the characteristics of electronic devices in various applications [62,63]. In the present study, we aimed to investigate whether this assumption holds true or not in the context of TENG energy harvesting technology. To support this hypothesis, we conducted analyses using Field Emission Scanning Electron Microscopy (FESEM) and Energy Dispersive Spectroscopy (EDS) on developed RC-WABs which were synthesized at various temperatures ranging from 0 to 800 °C as shown in Fig. 2. The corresponding results of it are presented in Figs. 2**a-2h** while Figs. 2**a1–2h1** demonstrate its highly magnified images. A noticeable transformation in the surface morphology of RC-WABs was observed in response to a change in the reaction temperature which is in agreement with previous reports that reaction temperature significantly affects the surface framework of the material [55,64–66]. When the reaction temperature ranged from 0 to 200 °C, aggregated particles were observed, as evident from Figs. 2**a1–2c1**. Subsequently, as the reaction temperature was increased from 200 to 300 °C, these aggregated particles started to exhibit a porous structure, as shown in Figs. 2**d1 and 2e1**. The desired highly mesoporous morphology of RC-WABs was achieved by maintaining the reaction temperature between 400 °C and 500 °C, as depicted in Figs. 2**e1 and 2f1**. Moreover, the mesoporous morphology transformed into highly micro-cubic crystals when the reaction temperature was set between 600 °C and 800 °C, as illustrated in Figs. 2**g1 and 2h1**.

This is because, during the pyrolysis of high temperature, several structural transformations and chemical reactions take place within the material [67,68]. It is well known that the thermal degradation process is a remarkably complex phenomenon and is known to involve a large number of elementary reactions and an interplay of solid, liquid, and gas-phase mechanisms [69–71]. Empirical evidence substantiates that higher temperatures induce intensified thermal fluctuations in atomic motion. This heightened movement allows molecules greater freedom and facilitates the breaking of chemical bonds, particularly those more susceptible to fracture [72,73]. Consequently, this phenomenon leads to a reduction in the ultimate strength of molecules, promoting either the transition to a crystalline state or an increased likelihood of fracture. However, it's crucial to note that while high temperatures promote crystallization, they also render materials more prone to structural degradation [74–76]. The pyrolysis process at high temperatures triggers complex structural transformations and chemical reactions within the materials. Therefore, as we mentioned above this thermal degradation process is intricate, involving numerous elementary reactions and an interplay of mechanisms across solid, liquid, and gas phases.

In our study, a similar mechanism was observed. At pyrolysis temperatures of 600 and 800 °C, an evident crystalline nature emerged within RC-WABs, as depicted in Figs. 2**g1 and 2h1**. This transformation arises from a thermal degradation process encompassing the removal or degradation of residual solvent molecules, rearrangement of polymer chains, and the establishment of well-defined crystal structures within C-WABs. These structural alterations are pivotal in comprehending the changes in surface morphologies observed at high-temperature ranges. Meanwhile, our findings align with already established reports that substantiate how material transformations occur within high-temperature ranges. These alterations result from a complex interplay of molecular rearrangements and crystalline formation induced by the thermal degradation process. This nuanced understanding sheds light on the observed changes in surface morphologies, underscoring the intricate nature of material behavior under pyrolysis conditions.

In addition to the FESEM analysis, we also performed Energy Dispersive X-ray Spectroscopy (EDS) on the RC-WABs samples using the same instrument equipped with an EDS detector. The corresponding EDS spectra are presented in Figs. 2**a2–2h2**. These spectra reveal variations in the concentration of carbon and oxygen atoms in the RC-WABs as a function of the reaction temperature. From Figs. 2**a2–2h2**, it can be observed that the carbon and oxygen atom concentrations differ depending on the reaction temperature. From the results, it can be seen that the pyrolysis of RC-WABs indicates that the carbon concentration gradually increases from 100 °C to 500 °C, reaching a pinnacle at a pyrolysis temperature of 500 °C and attaining a maximum carbon concentration of 50.12%, as depicted in Fig. 2f2. These findings offer valuable insights into the optimal pyrolysis conditions for the successful reactivation of RC-WABs in order to make exceptional tribopositive material for future TENG devices. At the same time, it was also observed that the carbon concentration reduces notably at higher temperatures (600 °C and 800 °C) as shown in Fig. 2g2 and 2h2 respectively. At the

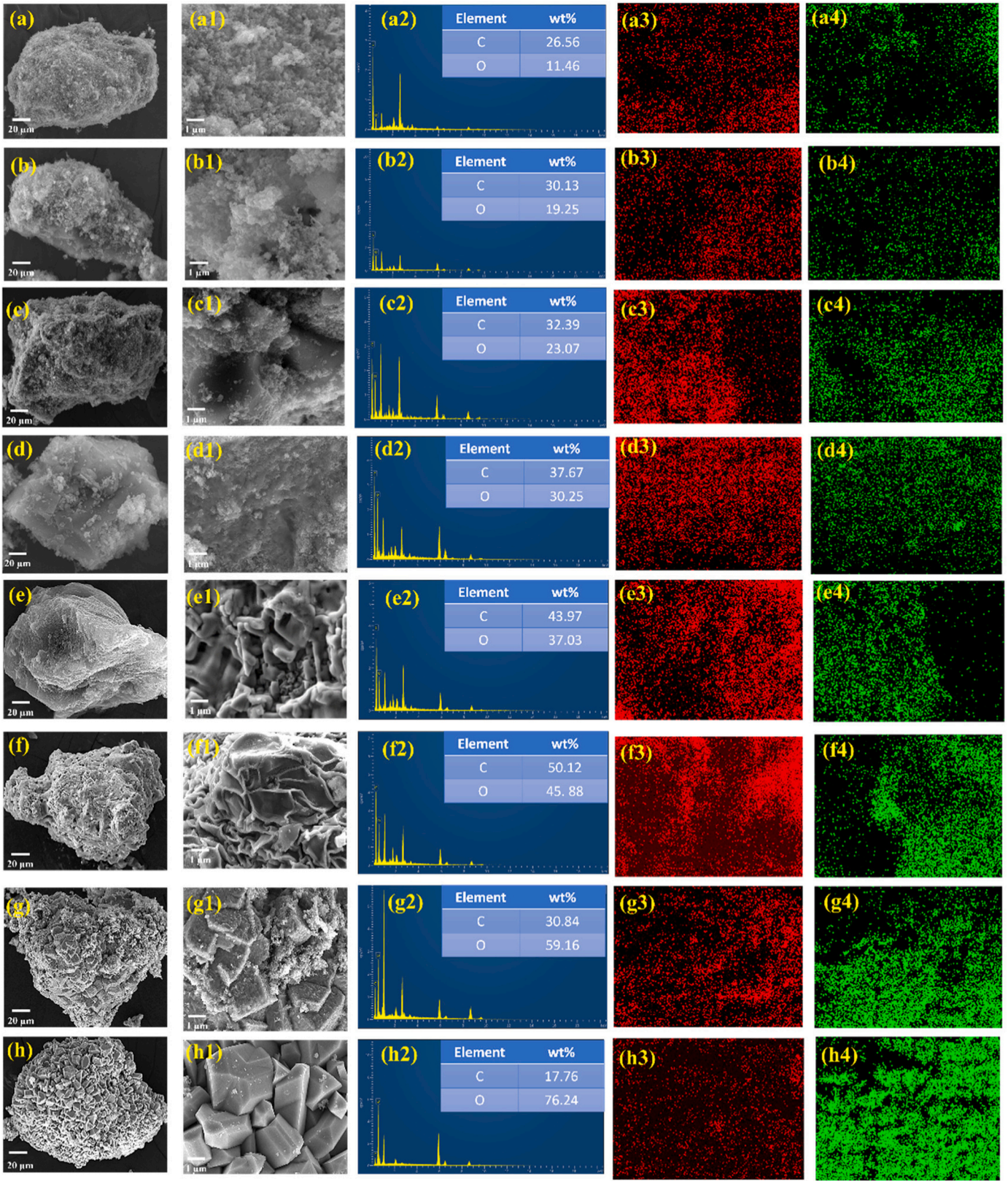


Fig. 2. : Morphological analysis of the investigated all RC-WABs of (a to h) at 2.00 kx magnification level and (a1 to h1) at 40.00 kx magnification level, respectively, with their corresponding elemental analysis as (a2 to h2). The weight percentages of each element are shown in the inset tables. The elemental distribution mapping uses an EDX detector for carbon in (a3 to h3) and oxygen in (a4 to h4).

maximum pyrolysis temperature of 800 °C, the carbon content within RC-WABs reached its nadir, registering a mere 17.55% atomic percentage (see Fig. 2h2). Meanwhile, there are several factors that are significantly responsible for the decline in the carbon content of C-WABs during pyrolysis above 500 °C without active gas (N_2, H_2, or Ar). These include intense thermal breakdown of carbon-rich compounds,

subsequent reactions with oxygen, resulting structural changes, evaporation of specific carbon elements, and the formation of volatile substances [77–86]. These combined mechanisms illustrate the complex changes happening within the material at high temperatures, resulting in a decrease in carbon concentration. Furthermore, Figs. 2**a3–2h3** and Figs. 2**a4–2h4** provide elemental mapping of the as-synthesized RC-WAB materials. These maps illustrate the spatial distribution of different elements within the samples.

To summarize, our analysis demonstrates that a reaction temperature of 500 °C is suitable for fabricating mesoporous and highly efficient tribopositive layered materials derived from the cathode material of WABs for TENG applications. This temperature yields RC-WABs with optimal carbon and oxygen concentrations, as evidenced by the EDS spectra (Fig. 2f2). Additionally, the elemental mapping images (Figs. 2**a3–2h3** and Figs. 2**a4–2h4**) provide further confirmation of the elemental distribution within the synthesized RC-WAB materials. This research highlights the feasibility of utilizing a reaction temperature of 500 °C to produce mesoporous RC-WABs, which hold great potential for enhancing the efficiency of TENG devices. Moreover, this approach offers a promising avenue for reactivating waste alkaline batteries (WABs) to create valuable materials for future TENG technology, contributing to sustainable development and resource utilization.

3.1.2. XRD analysis

X-ray diffraction (XRD) is a key technique used in science to study and understand the properties of materials. It helps us determine important characteristics like the lattice constants, crystallinity, crystal structure, and spacing of crystal planes in a material [87,88]. When we analyze crystalline materials using XRD, their atoms are arranged in a highly organized and orderly manner. This results in clear, sharp, and strong X-ray patterns, making it easy to identify the crystal structure and other properties [89–92]. In scientific terms, crystalline materials have a neat and organized atomic structure, which gives rise to sharp and strong X-ray patterns when studied with X-rays. On the other hand, microcrystalline materials have smaller and less orderly atomic regions, resulting in broader X-ray patterns. Amorphous materials, however, have a completely different atomic arrangement. They lack the regularity found in crystalline materials, making their X-ray patterns very broad and diffuse. This distinct pattern is quite challenging to interpret using the standard methods used for crystalline materials because amorphous materials do not exhibit the well-defined peaks characteristic of crystalline structures. Fig. 3**a** depicts a wide range XRD pattern of all RC-WABs materials synthesized under different reaction temperatures ranging from 0 to 800 ℃. We acknowledged each spectrum according to its synthesis temperature range, namely, RC-WABs-x (x = 0, 100, 200, 300, 400, 500, 600, and 800) as shown in Fig. 3**a**.

The peak at around 2θ = ~26° was observed in all spectrums (see the area under a yellow pattern) due to the carbon phase with mesoporous (JCPDS 00–056-0159) with the reflection in the (002) plane of aromatic layers [93–96]. While the peak centered at 2θ = ~45° (area under an orange pattern) which corresponds to the (100) plane of graphite carbon, can also be observed. The (002) peak is attributed to the orientation of the aromatic ring carbon reticulated layers in the three-dimensional arrangement [96,97]. Whereas the (100) peak is attributed to the amorphous state of carbonaceous materials [98,99]. The peaks at around 2θ = ~30, ~34°, and ~36° found in the XRD spectra of RC-WABs-bare, RC-WABs − 100, RC-WABs-200, and RC-WABs-300 (under the area of purple pattern) corresponds to the complex ionic species compositions such as Lix+ , Mnx+ , Fex+ , Cox+ , Px+ , and Ox+ , etc which were used to make cathode shell of ABs [100–103]. Remarkably, the XRD spectra of RC-WABs-400 and RC-WABs-500 no longer show these peaks that were present in the XRD spectra of RC-WABs-bare, RC-WABs − 100, RC-WABs-200, and RC-WABs-300. Instead, a new wider peak has appeared which indicates that the material has transformed from crystalline to mesoporous (see the yellow patterned inset in Fig. 3**a**). This might be due to the increase in pyrolysis reaction temperature. As it has already been reported that the pyrolysis temperature significantly affects the surface morphology of the materials [78,104]. Furthermore, in the XRD spectrum of RC -WABs-600 and RC-WABs-800, not only have all of the above peaks decreased but also new, sharp, and very intense peaks have appeared at about 2θ = ~46°, ~48° and ~52° (see the area under the bright green pattern), confirming the transformation into a different, new material, which is also supported by the change in the color of the material from black to brown, as shown in Fig. 3**a**. Overall, the RC-WABs active vacant sites have become more amorphous due to reaction temperature, which is advantageous for charge transfer from one electrode to another electrode, and the position and shape of the RC-WABs peaks have significantly changed, possibly contributing to the loss of the integrity of the crystal structure [105–108].

3.1.3. FTIR analysis

The application of Fourier Transform Infrared (FTIR) analysis is pivotal for unraveling the surface functional groups and chemical composition of materials. In our investigation, we scrutinized the influence of pyrolysis temperatures ranging from 0 to 800 °C on the surface functional groups and chemical composition of RC-WABs tribopositive materials derived from WABs. Fig. 3**b** illustrates the FTIR spectra, with nomenclature corresponding to the pyrolysis temperatures, such as RC-WABs-bare, RC-WABs-100, RC-WABs-200, and RC-WABs-300, RC-WABs-400, RC-WABs-500, RC-WABs-600, and RC-WABs-800. Notably, a conspicuous peak at 2115 cm^{-1} **(see** Fig. 3**b)** was consistently observed across all spectra, attributed to C═C stretching vibrations of the non-aromatic carboxylic group coupled with a keto group and the C-C-O bond, confirming the presence of carbon

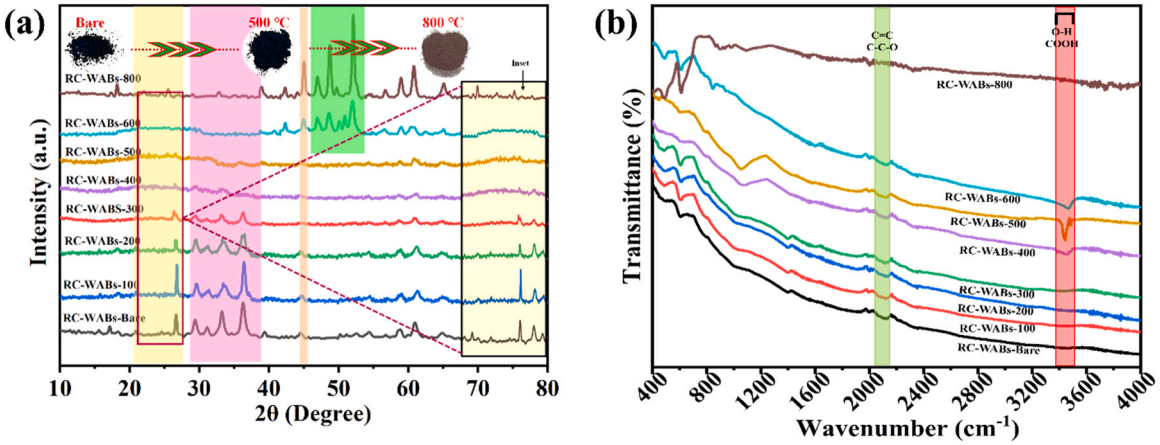


Fig. 3. : XRD (a) and the FTIR (b) spectra of all RC-WABs synthesized at different temperatures (0 to 800 ℃).

material [109]. However, in the FTIR spectrum of RC-WABs-600, a lower intensity of this peak compared to other spectra indicated a degradation of carbon concentration. Remarkably, this peak was entirely absent in the FTIR spectrum of RC-WABs-800, aligning with FESEM-EDS findings and signifying a substantial loss of carbon concentration, likely induced by the higher pyrolysis temperatures. Further scrutiny revealed an intriguing peak at approximately 3458 cm^{-1} in the FTIR spectrum of RC-WABs-400 and RC-WABs-500, absent in other spectrums (FTIR spectra of RC-WABs-bare, RC-WABs-100, RC-WABs-200, and RC-WABs-300, RC-WABs-600, RC-WABs-800) as shown in Fig. 3**b**. This peak corresponds to O-H/COOH stretching vibrations of hydroxyl groups [110,111]. According to our synthesis procedure, hydrolysis during pyrolysis induces a phase change in the ability of RC-WAB hydroxyl groups to form hydrogen bonds (H-bonds) [112,113]. These surface functional groups, encompassing O-H, COOH, C-O-C, and C-H, play a significant role in facilitating charge transformation between electrodes, as documented in the literature [114–116]. In conclusion, the comprehensive analysis from various

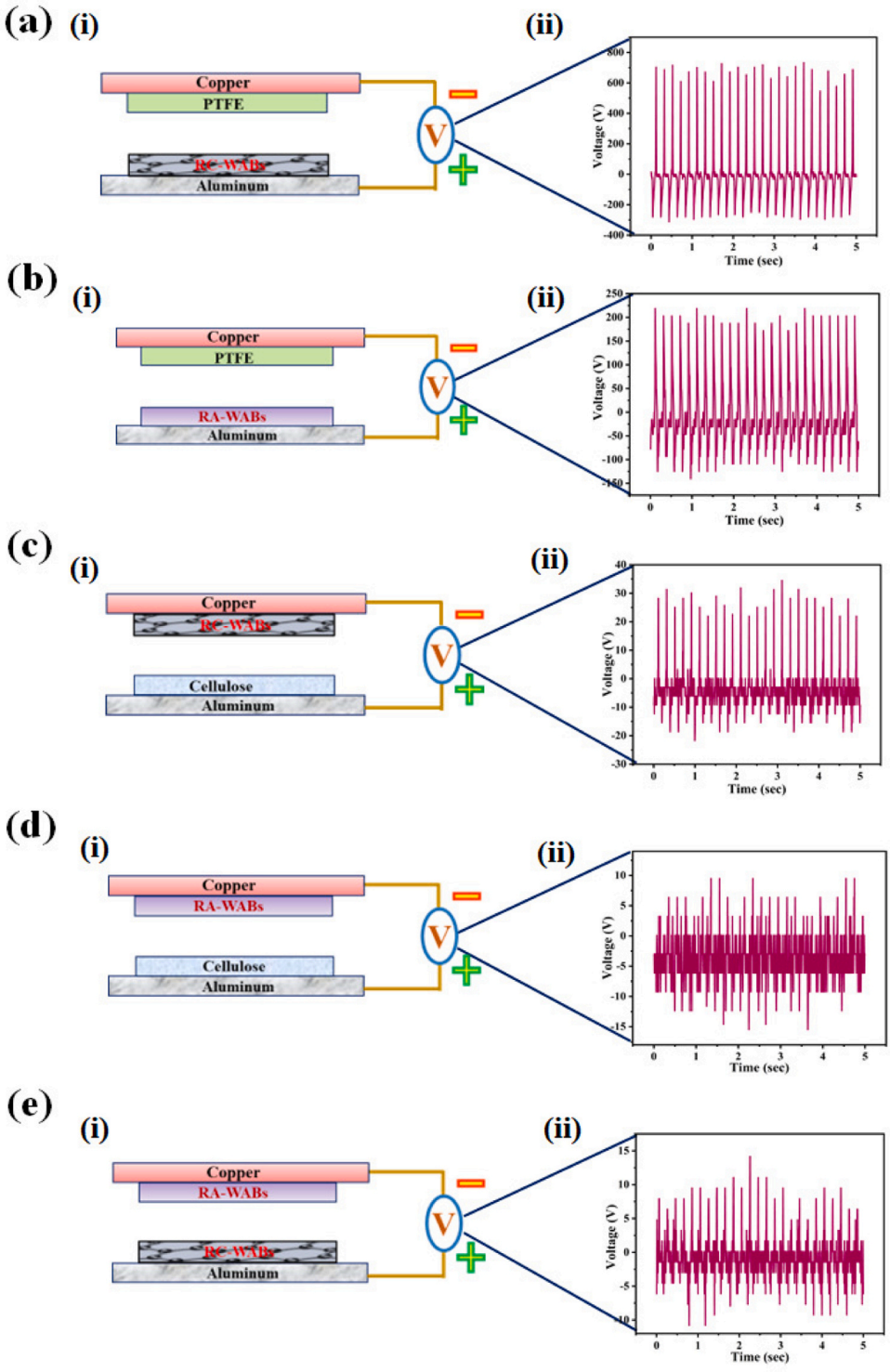


Fig. 4. Comparison of the output performance of RC-WABs and RA-WABs-based TENG devices in relation to the tribopositive and tribonegative layers. The structural representations of the fabricated devices with the different active layers as (ai, bi, ci, di, and ei), followed by their corresponding outputs as (aii, bii, cii, dii, and eii).

characterization techniques substantiates that 500 °C is the optimal pyrolysis temperature for the successful reactivation of C-WABs, rendering them exceptional tribopositive materials for future Triboelectric Nanogenerator (TENG) devices designed for large-scale energy harvesting applications. Additionally, our findings underscore the pronounced influence of reaction pyrolysis temperature on both the functional groups of the material surface and the material's surface morphology, providing valuable insights for the development of high-performance energy materials.

3.2. Electrical measurements

One of the primary objectives of our research is to address the challenge of waste management by transforming waste materials into valuable products. In pursuit of this goal, we have successfully developed a novel material that is not only affordable but also highly accessible for utilization in TENGs. This material is derived from waste alkaline batteries (WABs), which are considered universal wastes. By repurposing WABs, we aim to achieve both economic viabilities and provide a viable counter-electroactive layer for TENG applications. It is worth noting that our investigation represents a pioneering endeavor in the field, as we are the first to explore and employ WABs specifically for TENG applications. Through this innovative approach, we aim to contribute to the scientific community's understanding of waste utilization and its potential for generating value-added products. During this study, we reactivated and repurposed the electrodes (both cathode and anode) of WABs to serve as an electroactive layer for TENG energy harvesting. In order to find the best TENG device among these WAB-based TENG devices, we conducted an extensive examination of their output characteristics through a series of initial electrical measurements. To explore alternative electroactive layers that could potentially surpass the performance of RC-WABs (Reactivated Cathode-WABs) and RA-WABs (Reactivated Anode-WABs), we fabricated five distinct TENG devices as shown in Fig. 4. This figure provides a comparative analysis of the output performance of TENG devices based on different configurations. The performance is evaluated with respect to the utilization of RC-WABs and RA-WABs as tribopositive and tribonegative layers. Fig. 4 **(ai, bi, ci, di)** illustrates the schematic representations of fabrication processes of the respective TENG devices, followed by their corresponding output performance evaluations shown in Fig. 4 **(aii, bii, cii, and dii)** respectively. Fig. 4**(ei)** also includes a schematic representation of a TENG device based on RC-WABs as the tribopositive layer and RA-WABs as the tribonegative layer, along with its output performance analysis **(see** Fig. 4**(eii))**. These comprehensive visual representations provide valuable insights into the performance characteristics of different TENG device configurations, highlighting the role of electroactive layering arrangements. Analysis of the data presented in Fig. 4 reveals a distinct performance trend among the various TENG devices investigated. Particularly noteworthy is the exceptional performance exhibited by the TENG device incorporating RC-WABs as the tribopositive layer, surpassing the other TENG devices in terms of output performance. The supplementary movie S1 also highlights this difference. These findings indicate that RC-WABs hold significant promise as a tribopositive layer material for future TENG applications. To further explore our inquiries, we conducted an examination of the output performance of a TENG device utilizing cathode material from brand-new alkaline material (referred to as C-NABs), in comparison with our developed RC-WABs TENG device. supplementary Fig. S2 provided insights into the fabrication process of the TENG device based on C-NABs as tribopositive and tribonegative layers (see Fig. S2 **(ai, bi, ci))** while its corresponding output performance is shown in supplementary Fig. S2**(aii, bii, cii))** correspondingly. Upon analyzing supplementary Fig. S2, it is evident that the results demonstrate that the implementation of C-NABs as the cathode material did not yield satisfactory outcomes. The performance of the TENG device incorporating C-NABs exhibited lower efficiency when compared to the TENG device utilizing the RC-WABs tribopositive layer. This discrepancy is also highlighted in the supplementary movie S2. These findings provide scientific evidence that the incorporation of C-NABs as the cathode material did not lead to improved performance in the TENG device, in contrast to the more favorable outcomes achieved with the RC-WABs tribopositive layer. In addition to this, in our pursuit to understand the variations in output performance among TENG devices based on RC-WABs, RA-WABs, and the cathode material of brand-new alkaline batteries (C-NABs), we conducted an in-depth analysis as shown in supplementary Fig. S3. Detailed information about them is provided in the supplementary information **note 1**. Based on the observed superiority of RC-WABs, we have decided to adopt them as the fixed tribopositive layer material for further investigation of the second assumption in this study. This decision is supported by the compelling evidence obtained from our experimental results, which highlight the potential of RC-WABs to enhance the overall performance of TENG devices. By utilizing RC-WABs as a consistent tribopositive layer, we aim to explore and validate the second assumption underlying this research endeavor.

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To address the second objective of our study, we conducted a subsequent series of electrical measurements to investigate the impact of surface morphology and oxygen content availability on the output performance of the TENG devices. Specifically, we sought to determine whether these factors influenced the efficiency and functionality of the TENGs. By carefully analyzing the surface morphology of the TENG devices, we aimed to discern any correlations between specific structural characteristics and the observed output performance. Additionally, we examined the availability of oxygen content within the TENG devices to assess its potential influence on the generated output. These electrical measurements were crucial in providing valuable insights into the relationship between surface morphology, oxygen content availability, and the performance of the TENG devices. By employing rigorous scientific methodologies and precise data analysis techniques, we aimed to ascertain whether these factors have a significant impact on the overall output performance of the TENGs. For that, we assembled 7 different TENG devices based on their variation in synthesis temperature as shown in supplementary Fig. S4. We assigned them designations like those used in the XRD, namely RC-WABs-x (x = 0, 100, 200, 300, 400, 500, 600, and 800 °C). The **SI** Table S1 provides detailed information about device fabrication parameters and experimental conditions for each TENG device based on the RC-WABs tribopositive layer. Each TENG device has been a 4 × 4 cm^2 size and is made in the same way. Each device has been subjected to an external force of 5 Hz. In this study, we tested all the TENGs and their corresponding output performance such as voltage and current.

Previous research has shown that the surface frameworks of materials change with reaction temperature [64,65,117]. As the reaction temperature increased, the porous structure of RC-WABs appeared to form. This has been presented in the corresponding FESEM results. Furthermore, the output performance of the RC-WABs-based TENG device was found to be affected due to variations in the RC-WABs surface morphology and their availability of oxygen content, as shown in Figs. 5**a and** 5**b**. The performance of the TENG device based on RC-WABs increased up to the reaction temperature of 500 °C. After 500 °C, the performance started decreasing at higher temperatures of 600 °C and 800 °C, as depicted in Figs. 5**a and** 5**b**. Therefore, changes in surface morphologies and oxygen content of RC-WABs-based TENG devices occur because of varying reaction temperatures [118–123]. The TENG device utilizing RC-WABs-500 demonstrated exceptional output performance compared to the other TENG devices (RC-WABs-bare, RC-WABs-100, RC-WABs-200, RC-WABs-300, RC-WABs-400, RC-WABs-600, and RC-WABs-800). This indicates that the reaction temperature of 500 °C is the optimal condition for obtaining a highly porous tribopositive electrode material from WABs. This aligns with the findings obtained through various characterization techniques (FESEM, XRD, and FTIR as well). The highly porous nature and favorable oxygen content availability in RC-WABs-500 are likely attributed to this optimal

temperature choice. Previous studies have highlighted the significance of surface roughness and oxygen availability in enhancing the tribo-electrification process and improving the output performance of TENG device [50,124–126]. The favorable oxygen content and rough surface of RC-WAB-500 are believed to be contributing factors to the enhanced output efficiency observed in the RC-WAB-500-based TENG device. These findings support the hypothesis that the presence of a rough surface and reasonable availability of oxygen play crucial roles in enhancing the output efficiency of TENG devices. The optimized reaction temperature of 500 °C for RC-WABs provides an ideal basis for fabricating highly porous tribopositive electrodes. This is supported by the comprehensive characterization results. These results further emphasize the importance of oxygen content and surface morphology in developing efficient TENG devices. Thus, contributing to the scientific understanding and advancement of TENG technology [54,123,127].

As a continuation of the previous predictions, another successful experiment was conducted. The waste cathode material was subjected to pyrolysis within a tube furnace, utilizing controlled atmospheric conditions with an N_2 atmosphere. The experimental parameters, such as reaction temperature and reaction time, remained constant throughout the process. After the reaction was complete, the furnace was allowed to cool naturally to reach room temperature. The resulting material was then finely crushed into a powder known as TF-RC-WABs synthesized in a tube furnace using a mortar and pestle. This powder was then examined as a tribopositive layer material, like other RC-WABs electropositive layers, and the results are presented in supplementary Fig. S5. From supplementary Fig. S5, it is found that the TENG device based on the TF-RC-WABs tribopositive layer exhibited lower electrical output as compared to the TENG device based on the RC-WABs tribopositive layer. Experimental results were gently demonstrated in the supplementary movie S3. This disparity in performance may be attributed to the reduced availability of oxygen content within the TF-RC-WABs tribopositive layer. As previously mentioned, synthesis under an inert atmosphere reduces the presence of hydroxyl groups on the material's surface [128–131].

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As it is well known TENGs are energy harvesting devices that rely on the triboelectric effect, which is based on the friction between two electroactive layers viz., first one is tribopositive and another is tribonegative [132]. Meanwhile, as previously stated, we used the prepared RC-WABs electrode material in this work as a tribopositive electrode material. However, in the second set of experiments, we investigated the impact of different electronegative layers on output performance of RC-WABs-500-based TENG devices. For that we altered the electronegative layer with cellulose, lignin, PET, and PTFE. The corresponding output performances (both voltage and current) of effect of different electroactive layers are shown in Figs. 5**c and** 5**d**, respectively. From Figs. 5**c and** 5**d**, it is observed that the output performance of the RC-WABs-500-PTFE TENG device shows exceptional output performance as compared to other TENG devices. However, according to the findings herein we fixed PTFE as an electronegative layer for the RC-WABs-500 TENG device and employed the further remaining experiments [65].

Besides, in the next set of experiments, we measured the detailed electrical outputs of the proposed RC-WABs-based TENG device. In this experiment, we have performed experiments with and without RC-WABs tribopositive layer and its corresponding results are pointed out in Figs. 6**a and** 6**b**, respectively. Figs. 6**a and** 6**b** show that without the RC-WABs tribopositive layer, there is no output performance (both

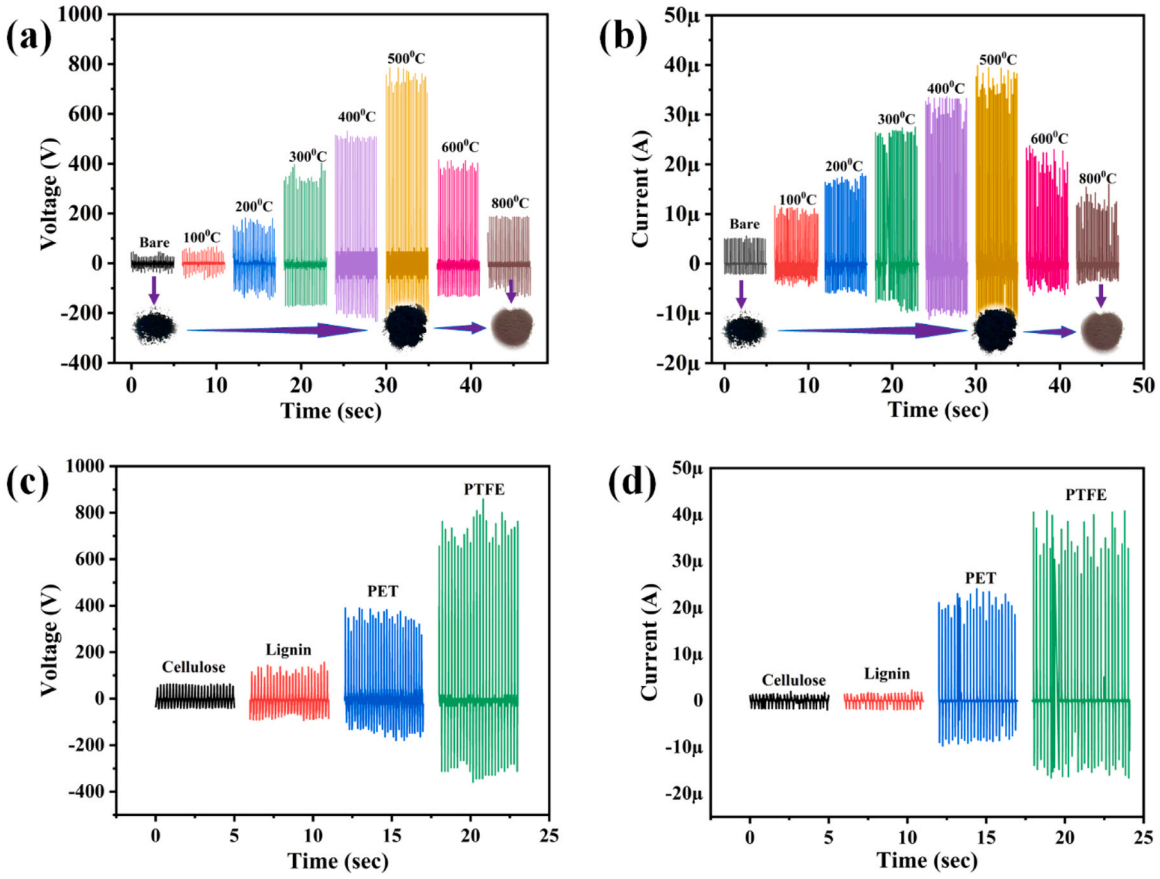


Fig. 5. : Comparative analysis of output voltage (a) and the current (b) characteristics of different TENG devices based on the various RC-WABs materials synthesized at various synthetic temperature conditions (0 to 800 °C). Comparative analysis of output voltage (c) and the output current (d) of the RC-WABs-500 based TENG device using different tribo-negative layered materials such as cellulose, lignin, PET, and PTFE.

voltage and current) whereas, with RC-WABs tribopositive layer the output performance was increased abruptly. It was jumped up from ~51 V to ~775 V and ~1.5 μA to ~39.13 μA respectively, which indicates that the RC-WABs based TENG devices showing excellent output performance. Further to test the reproducibility of the proposed device many experiments were performed which yielded the same outcomes.

The working conditions must be improved to evaluate the TENG devices' reasonable output performance and check their scientific foundation. However, there are three crucial factors such as frequency, applied power, and contact area that significantly affected the overall performance of every TENG device. As a result, herein in this section, we tested the effect of various frequencies on our proposed RC-WABs-500-based TENG device. In this study, we conducted experiments by altering the applied frequencies of the TENG devices. Specifically, we tested frequencies of 2 Hz, 3 Hz, and 5 Hz, and their corresponding output performance was recorded and analyzed. The results are presented in Figs. 6**c and** 6**d**, which depict the output voltage and current, respectively. From our observations in Figs. 6**c and** 6**d**, a clear trend emerges. As the frequencies increased from 2 Hz to 5 Hz, the output voltage and current exhibited a steady rise, reaching approximately ~374 V to ~786 V and ~11 μA to ~39.17 μA, respectively. This increase in output performance can be attributed to the effect of higher frequencies on the TENG device. As the working frequency is increased, the electrode layers come into contact more frequently, resulting in the generation of more charges. This phenomenon directly contributes to the increase in output voltage and current of the TENG device. Based on these findings, we propose that

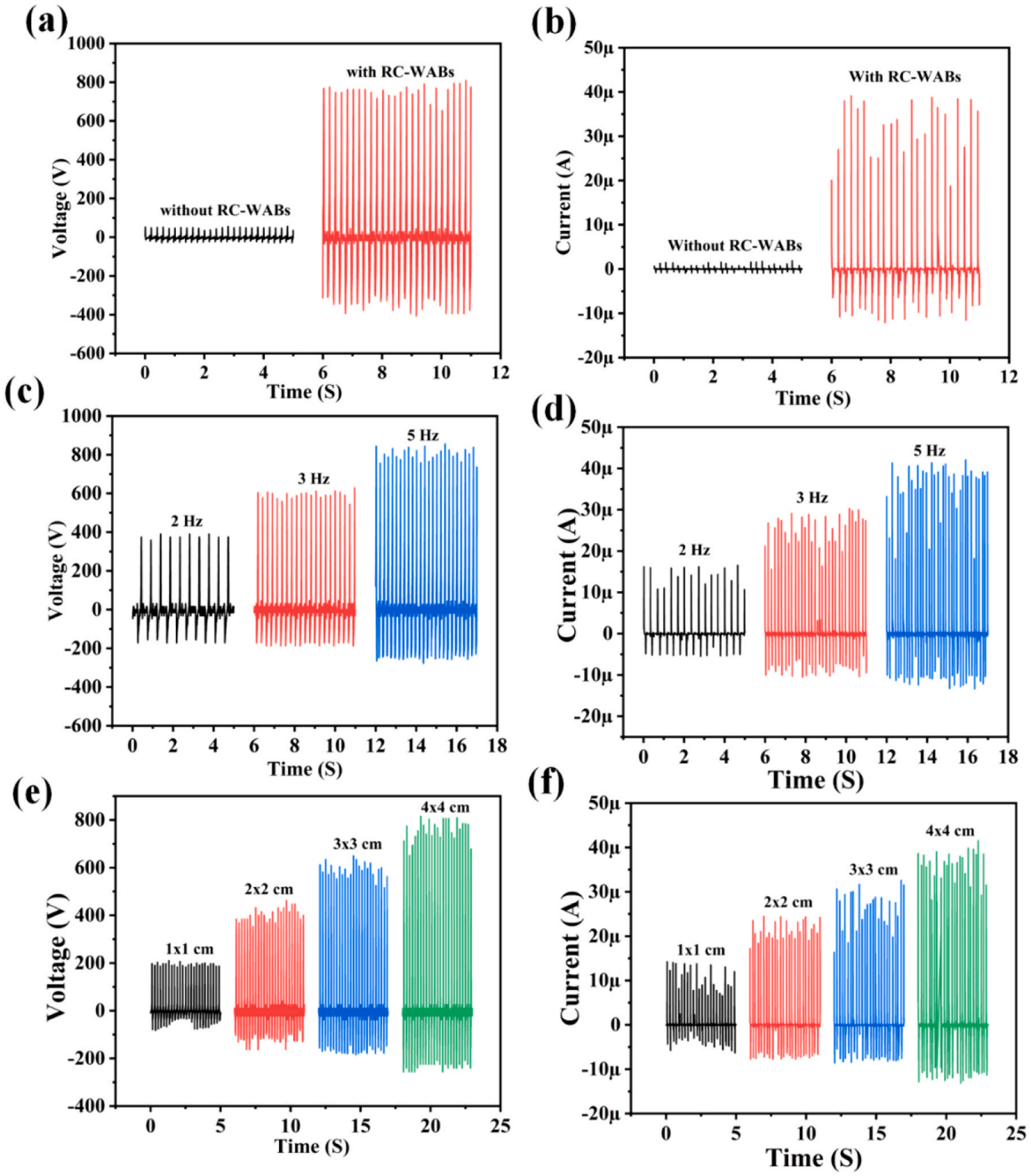


Fig. 6. : Output voltage (a) and current (b) characteristics of RC-WAB-500-based TENG devices with and without RC-WAB-500 tribopositive layers. Output voltage (c) and current (d) of the RC-WABs-500-based TENG device at different frequencies (2 Hz, 3 Hz, and 5 Hz). The effect of different device sizes on the output voltage (e) and current (f) of the RC-WAB-500-based TENG device.

TENGs can be utilized in a wide range of scenarios involving different human behaviors. The ability of TENGs to generate higher output performance at increased frequencies suggests their suitability for diverse applications. By incorporating TENGs into various circumstances, we can harness the benefits of enhanced energy generation and utilization, thus paving the way for the adoption of TENGs in practical settings. In summary, this experiment provides scientific evidence indicating that increasing the frequency of TENG devices leads to an improvement in output voltage and current. This finding highlights the potential of TENGs for a broad range of applications, making them a promising technology for harvesting energy from human activities. Likewise, we also explored the influence of surface shapes on the output performance of the TENG device. We maintained the device sizes at $1 \times 1\ cm^2$, $2 \times 2\ cm^2$, $3 \times 3\ cm^2$, and $4 \times 4\ cm^2$ while keeping other experimental parameters such as applied frequency and force constant. The results are presented in Figs. 6**e and** 6**f**, illustrating the output voltage and current of the TENG devices, respectively. From the observations in Figs. 6**e and** 6**f**, a notable trend emerges. As the device size increases, the output performance of the TENG devices, in terms of both voltage and current, also increases. This phenomenon can be attributed to the larger contact surfaces of the TENG as the device size increases. The larger contact surfaces facilitate a more significant interaction between the electroactive layers, leading to a higher generation of electrical potential. Consequently, the TENG device exhibits improved output performance regarding voltage and current. These findings provide scientific evidence to support the understanding that larger surface shapes of TENG devices contribute to enhanced output performance. By expanding the device size, we can effectively increase the contact area, thereby maximizing the potential for energy generation. This can be attributed to the larger contact surfaces, which facilitate a more efficient generation of electrical potential. These findings have implications for the practical application of TENG technology, as utilizing larger surface shapes can help optimize energy harvesting and utilization in various scenarios.

Further, to verify that such throughput output performance is achieved by the suggested RC-WABs-500-based TENG device rather than a variety of external sources, we performed polarization switching tests on Fig. 7**a** (polarity switching test related to output voltage) and **7b** (polarity switching test related to output current). The output performance (voltage and current) is the same for reverse and forward connections as shown in Figs. 7**a and** 7**b**. So far, it has been confirmed that the output performance of the RC-WABs-based TENG system is generated exclusively by the TENG system and does not depend on external factors. The AC signals generated by the TENG device were rectified to convert them into a suitable form for powering low-power electrical devices. This rectification process involves converting the AC signals into direct current (DC) signals. The rectified DC signal was then utilized to charge three different commercially available capacitors: 0.1 μF, 4.7 μF, and 10 μF. Fig. 7c represents the capacitor charging characteristics when connected to the TENG device consisting of the RC-WABs-based tribopositive material. During the experiment, the TENG device was subjected to continuous squeezing at a frequency of 5 Hz, ensuring consistent and repetitive pressure application. The results depicted in Fig. 7**c** demonstrate the rapid charging performance of the TENG device, as all the capacitors reached higher voltages in a short period of time. This indicates the device's ability to efficiently transfer power to low-powered electronic devices. By delivering higher voltages, the RC-WABs tribopositive material enables enhanced power output, making it suitable for driving and operating various low-power electronic devices.

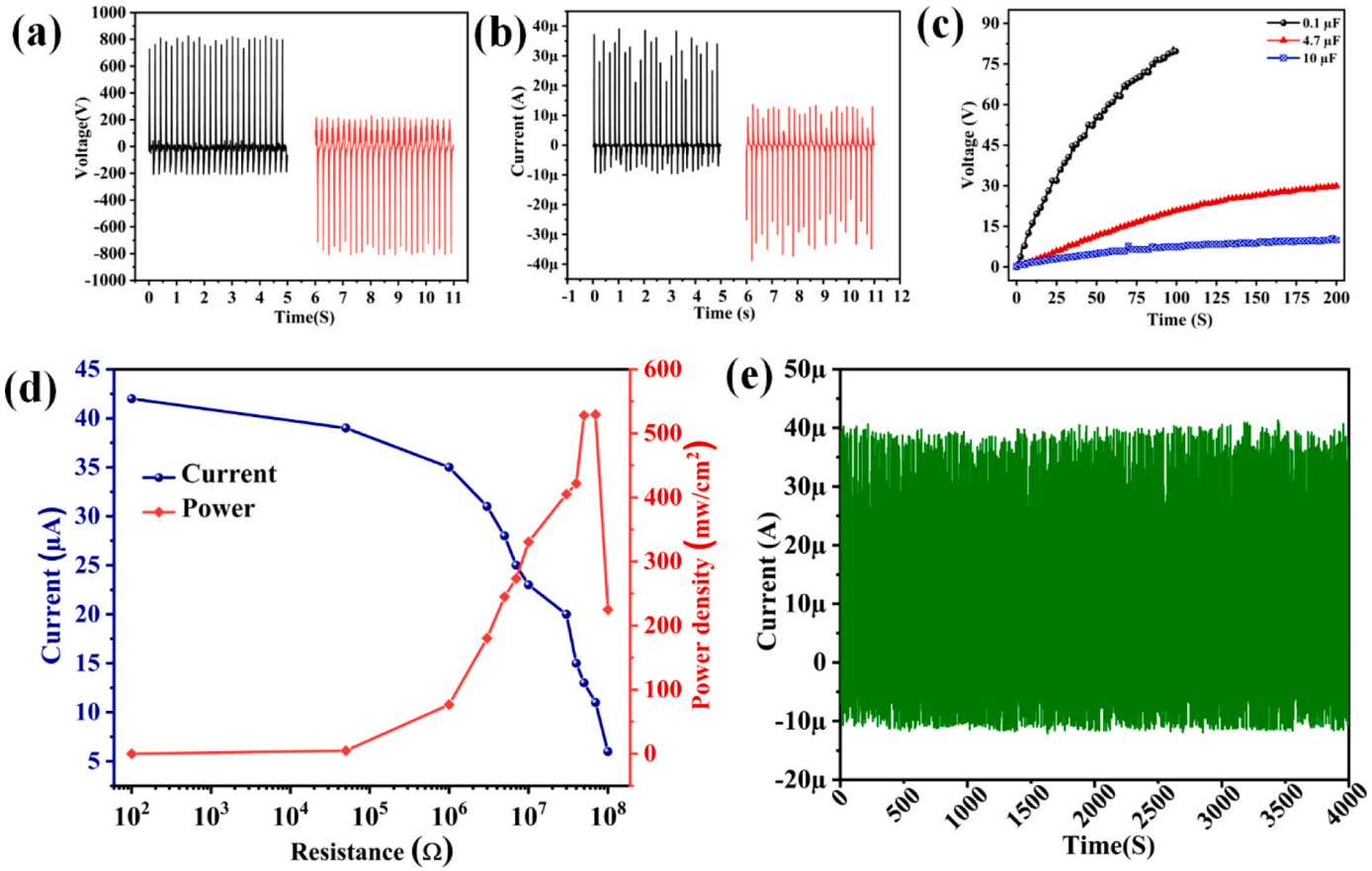


Fig. 7. : Forward and reverse output voltage (a) and the current (b) characteristics of the RC-WAB-500-based TENG device. (c) Charge-transfer properties of the TENG device at various commercially available capacitors (1, 4.7, and 10 μF). (d) Power density and output current with varying load resistance. The stability of the proposed RC-WABs-500-based TENG device for 40, 000 cycles (e).

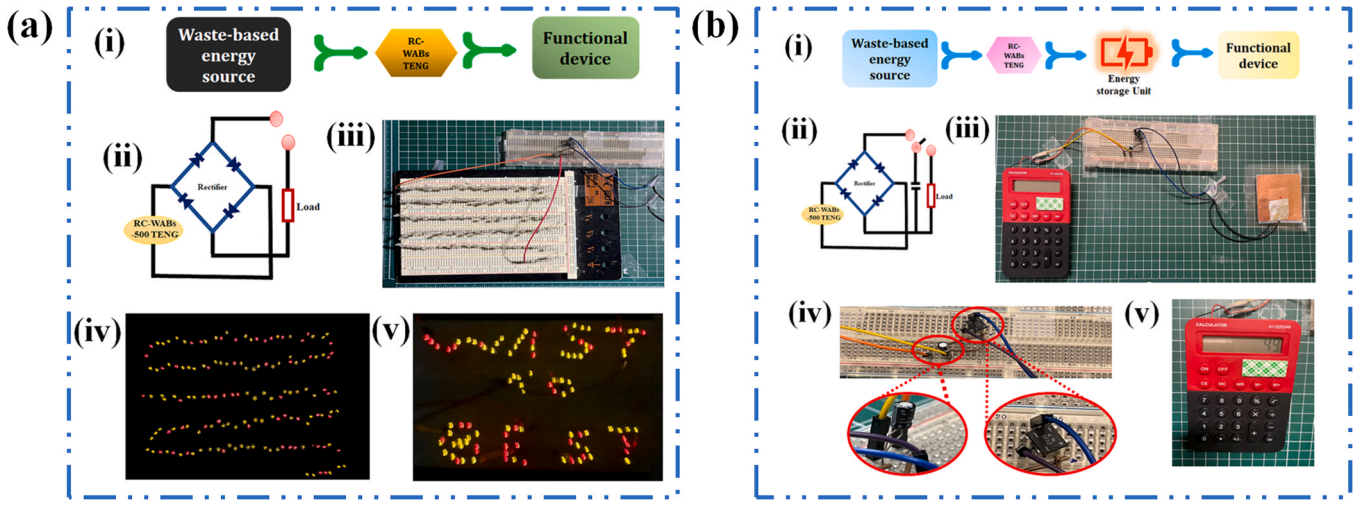


Fig. 8. The practicality of the TENG device based on the RC-WABs-500 as a power source for LED glowing (a) and powering the calculator (b). The system diagram (a (i)) and Circuit are designed to light up the commercial LEDs (a(ii)). LEDs in series with the RC-WABs-500-based TENG through a bridge rectifier (a(iii)). RC-WABs-500-based TENG as a powering source to light up 150 commercial LEDs (a(iv)), and an abbreviation "WASTE TO BEST" (a(v)). The system diagram (b(i)), and circuit diagram to power up the electronic calculator (b(ii)). Connecting the electronic calculator with the RC-WABs-500-based TENG through a 10 μF capacitor and bridge rectifier (b(iii)). Connection of the RC-WABs-500-based TENG with an electronic calculator (b(iv)). Operation of the electronic calculator (b(v)).

In addition to the findings, an assessment of the RC-WABs-based TENG device's instantaneous power and consistency was conducted to determine its power output and operational duration at full capacity. Fig. 7**d** presents the power output characteristics of the device when subjected to varying load resistors ranging from 100 Ω to 100 MΩ. The results indicate an inverse relationship between the load resistance and the current output, with higher resistances resulting in lower currents. Notably, the RC-WABs-based TENG device exhibited a significant power density of approximately 529.375 μW/cm^2 at a load resistor of approximately 70 MΩ (see Fig. 7**d**), surpassing the performance of previously reported waste-based TENG devices. This demonstrates the improved power generation capabilities of the hypothetical RC-WABs-based TENG. To provide a comprehensive understanding, the **SI** Table S2 offers a comparative analysis of the output performances of the fabricated RC-WABs-based TENG devices with previously reported waste-based TENG devices. The results highlight the superior power generation potential of the RC-WABs-based TENG. Additionally, the research also addressed the device's reliability, durability, and dependability by evaluating its stability over extended operating cycles. Fig. 7**e** illustrates the consistent output current of the RC-WABs-based TENG device, which remained unchanged even after 40,000 cycles. This signifies the device's high stability and durability under prolonged operation, further validating its potential for practical applications.

The study aimed to investigate the effects of external environmental factors, specifically humidity and temperature, on the efficiency of TENG devices. Previous studies have shown that TENG efficiency is significantly affected by changes in humidity and temperature [111,133–135]. Researchers have found that using waste materials as the electroactive layer in TENG construction leads to a notable decrease in performance under varying humidity and temperature conditions. In this study, we proposed a new waste material, RC-WABs, as a tribopositive layer in TENG technology. This marks the first instance of its utilization in this context. To evaluate the viability and functionality of the proposed TENG device integrating RC-WABs, a series of experiments were conducted. The final set of experiments focused on evaluating how changes in humidity and temperature impacted the device's output performance. These investigations are crucial in determining the practical applicability and efficacy of the RC-WABs-integrated TENG device. The results of the experiments, including methodologies and outcomes, have been systematically organized in note 2 of supplementary information**.** This cataloging provides valuable insights into the behavior and performance of the RC-WABs-based TENG device under varying environmental conditions, which is essential for assessing its real-world potential.

3.3. *Applications of RC-WABs-based TENG*

To explore the feasibility of producing self-powering systems using RC-WABs-based TENG devices, we investigated. Fig. 8**a(i)** provides an overview of the designed setup for the self-powering technology based on RC-WABs-TENG, which was utilized to power low-power electronics. For this experiment, we have maintained the same dimensions that we followed in the whole experiments, such as device size, electroactive layered thickness, and electrode spacing with the contact separation (CS) mode of our proposed RC-WAB TENG device. Additionally, Fig. 8**a (ii)** presents the equivalent circuit design for this system. To assess the practical implementation of RC-WABs-based TENG devices, we connected 150 LEDs in series to the TENG device, as depicted in Fig. 8**a(iii)**. Remarkably, the TENG device effectively drove these commercial LEDs, illuminating them to display the phrase ***"WASTE TO BEST"*** as illustrated in Fig. 8**a(iv) and** Fig. 8**a(v)** respectively. This demonstrates that the TENG built on the RC-WABs platform can function as both a power source to directly power electronics and to charge power storage devices. To create a self-powered system, it can be used in conjunction with an energy storage unit, as depicted in Fig. 8**b(i)**. The equivalent circuit for this system is illustrated in Fig. 8**b(ii)**, while Fig. 8**b(iii)** shows the schematic diagram of the system, with a capacitor serving as the storage device. In our experiments, we employed a 10 μF capacitor to run low-power electronic devices, such as a calculator. Fig. 8**b(iv)** showcases the connection of the RC-WABs-based TENG device to the calculator, and Fig. 8**b(v)** demonstrates that our proposed device can efficiently power the commercial calculator. Furthermore, we have provided supplementary movie S4 to showcase real-time applications of the aforementioned devices, providing a visual demonstration of their performance and capabilities. These scientific demonstrations highlight the potential of RC-WABs-based TENG devices as self-powering solutions for low-power electronics. They not only directly power devices but can also charge energy storage devices, enabling continuous and sustainable operation. The practical applications of this technology are vast, and the successful demonstration of powering commercial LEDs and a calculator underscores its efficacy. Please refer to supplementary movie S4 for a more visual representation of these devices in action.

Supplementary material related to this article can be found online at doi:10.1016/j.nanoen.2023.109205.

4. Conclusions

This paper has focused on the development of a waste-based material with high potential for use in TENG energy harvesting technology. The material was derived from RC-WABs through a simple and efficient pyrolysis process. Notably, this experiment was conducted at low temperatures without the need for activating gases such as N_2, H_2, or Ar_2, in an ambient temperature environment. To verify the successful reactivation of the cathode material of WABs, various characterization techniques were employed, including XRD, SEM, and FTIR. These techniques confirmed the reactivation of the cathode material of WABs and revealed significant changes in the surface morphology and surface functional groups of active oxygen based on the varying reactivation temperatures (ranging from 0 °C to 800 °C). Based on the different reactivation temperatures and the corresponding surface morphologies studied, TENG devices were fabricated using PTFE as the tribonegative layer and the developed RC-WABs as the tribopositive layer. The results indicated that the TENG device based on the reactivation temperature of 500 °C (RC-WABs-500) exhibited superior performance compared to other reactivation temperatures. This device (RC-WABs-500) demonstrated remarkable output voltage, current, and power density values of 790 V, 40.27 μA, and 529.375 μW/cm^2, respectively. This superior performance can be attributed to the relatively rough surface and high availability of oxygen content in the material, which enhances the triboelectric effect in TENG devices. To evaluate the stability of the developed material, it was subjected to 40,000 consecutive cycles, and the results demonstrated good stability over this duration. Furthermore, the material was successfully tested in practical applications such as powering LEDs and operating a digital calculator. These findings not only contribute to the recycling of waste alkaline batteries but also provide a promising pathway for the development of efficient and environmentally friendly materials for TENG energy harvesting. This research contributes to the sustainable advancement of this emerging field.

CRediT authorship contribution statement

Bae Jinho: Conceptualization, Formal analysis, Funding acquisition, Supervision, Writing – original draft, Writing – review & editing. **Dongle Tukaram D.:** Formal analysis, Writing – original draft. **Patil Swapnil R.:** Formal analysis, Writing – review & editing. **Noman Muhammad:** Formal analysis, Writing – review & editing. **Patil Chandrashekhar S.:** Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Saqib Qazi Muhammad:** Formal analysis, Writing – original draft, Writing – review & editing. **Kim Jungmin:** Formal analysis, Investigation. **Ko Yongbin:** Formal analysis, Investigation. **Chougale Mahesh Y.:** Formal analysis, Methodology. **Shaukat Rayyan Ali:** Formal analysis, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

Acknowledgments

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.nanoen.2023.109205.

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